SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Order Dispatch & Delivery Tracking System

38_TransitCoders

Team Members :

Vinod Katevarapu (S20240010256)

Jay Suriya R (S20240010093)

Mohammad Shazil A M (S20240010144)

Prasanna Raja R (S20240010191)

Shreesaanth R (S20240010224)

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Introduction

The Order Dispatch & Delivery Tracking System is a web-based application developed to manage the delivery process from order creation to final delivery confirmation. It helps different users such as merchants, hub managers, delivery agents, and customers coordinate delivery operations through a single platform. The system follows secure, rule-based logistics operations to ensure proper order handling, tracking, and confirmation at each stage.

System Objectives

The objective of this project is to develop an Order Dispatch & Delivery Tracking System that manages deliveries from order creation to final confirmation. It supports real-world logistics workflows such as order management, agent assignment, and delivery tracking. The system ensures secure access for different users and is designed to support future improvements.

Users Classes

Merchant	–	Creates and manages delivery orders
Hub Manager	–	Assigns delivery agents and monitors deliveries
Delivery Agent	–	Executes deliveries and updates order status
Customer	–	Tracks orders and confirms delivery
Administrator	–	Manages system configuration and business rules

Functional Requirements

User Requirements

- Merchant :
    - Schedule bulk deliveries and arrange for pickups.
    - Calculate price estimates and fill in detailed product information.
    - Generate unique Barcodes or QR codes for order identification.
- Hub Manager:
    - monitor products across different hubs and contact other hub managers.
    - They have the authority to manually override delivery assignments and contact drivers directly.
    - They can initiate "Send to different hub" commands for logistics routing.
- Delivery Agent:
    - Scan packages to instantly retrieve customer and delivery details.
    - Update the status of a delivery, including reporting mishaps or unsuccessful attempts.
    - View their assigned worklist and delivery history.
- Customer:
    - Sender:Calculate shipping prices and schedule a pickup.
    - Receiver: Access full real-time tracking of their package.

System Requirements

- Authentication**:**

  The system shall maintain a secure authentication gate to manage access for all four actor types.

- Inventory Management:

  The system shall provide a Warehouse Management System interface for hub organization.

- Notification Logic:

  The system shall automatically notify Local Delivery Partners and generate worklists when orders reach their territory.

- Tracking Engine:

  The system shall process GPS data to provide the status for the customer interface.

- Smart Routing Algorithm**:**

  The system shall automatically calculate the most fuel-efficient and time-effective route for delivery agents using algorithms.

- Concurrency Support:

  The system shall handle concurrent users by utilizing database connection pooling.

- Dispatch Automation:

  The system shall automatically assign orders to available delivery partners based on location and workload, with manual override support.

Non Functional Requirements

- Performance**:**

  The routing algorithm must generate an optimized path.

- Accuracy:

  The system must integrate with a mapping API (like Google Maps) to ensure route data is based on live traffic conditions.

- Scalability**:**

  The system must handle simultaneous routing requests from multiple hubs as the network grows.

- Security:

  Role-based access control must be enforced for Admin, Hub Manager, Delivery Partner, Merchant, and Customer and sensitive data such as user credentials and payments must be securely stored.

UML Models and Diagrams

This section presents UML diagrams that show the system's workflows, actor interactions, and overall operation flow.

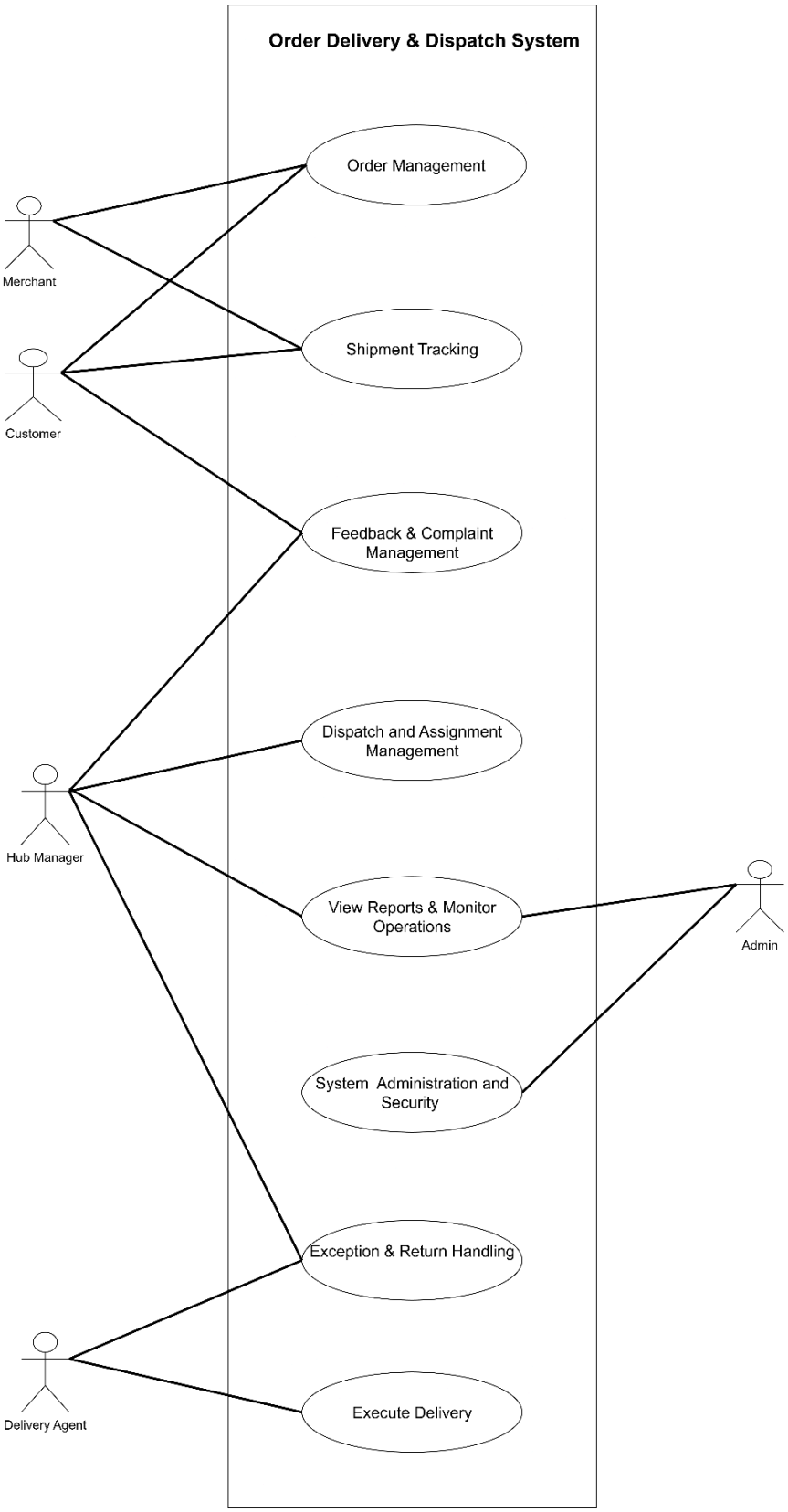
Use Case Diagram

The Use Case Diagram illustrates the interactions between the system and its actors. It identifies the major functionalities provided by the system and maps them to the respective actors.

The Use Case Diagram covers core functionalities such as:

- Order Management
- Dispatch & Assignment Management
- Delivery Execution
- Exception & Return Handling
- Shipment Tracking & Notification
- Feedback & Complaint Management
- Reporting & Monitoring
- System Administration & Security

USE CASE


Order Delivery & Dispatch System
Order Management
Shipment Tracking
Feedback & Complaint Management
Dispatch and Assignment Management
View Reports & Monitor Operations
System Administration and Security
Exception & Return Handling
Execute Delivery
Merchant
Customer
Hub Manager
Admin
Delivery Agent

Activity Diagrams

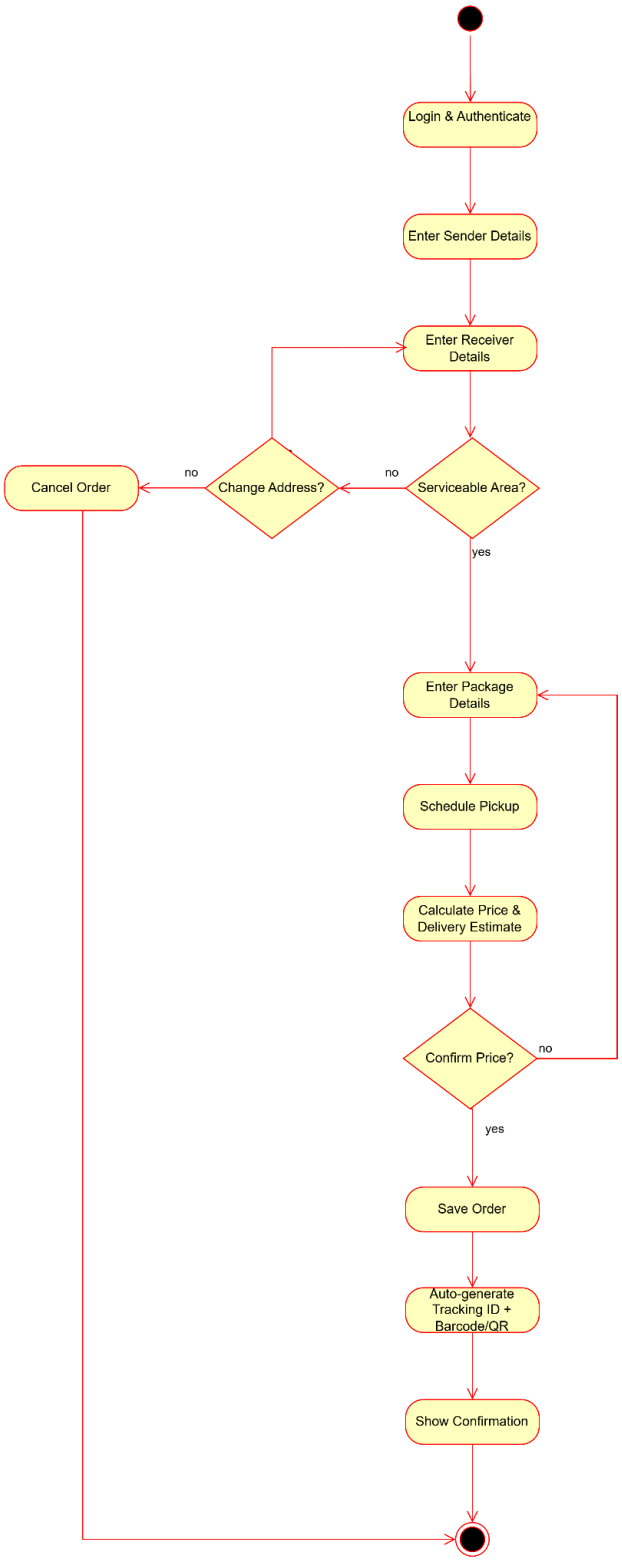
The Activity Diagrams describe the detailed workflows of major system processes. These diagrams represent step-by-step execution of tasks and decision points involved in logistics operations.

Key workflows modeled using Activity Diagrams include:

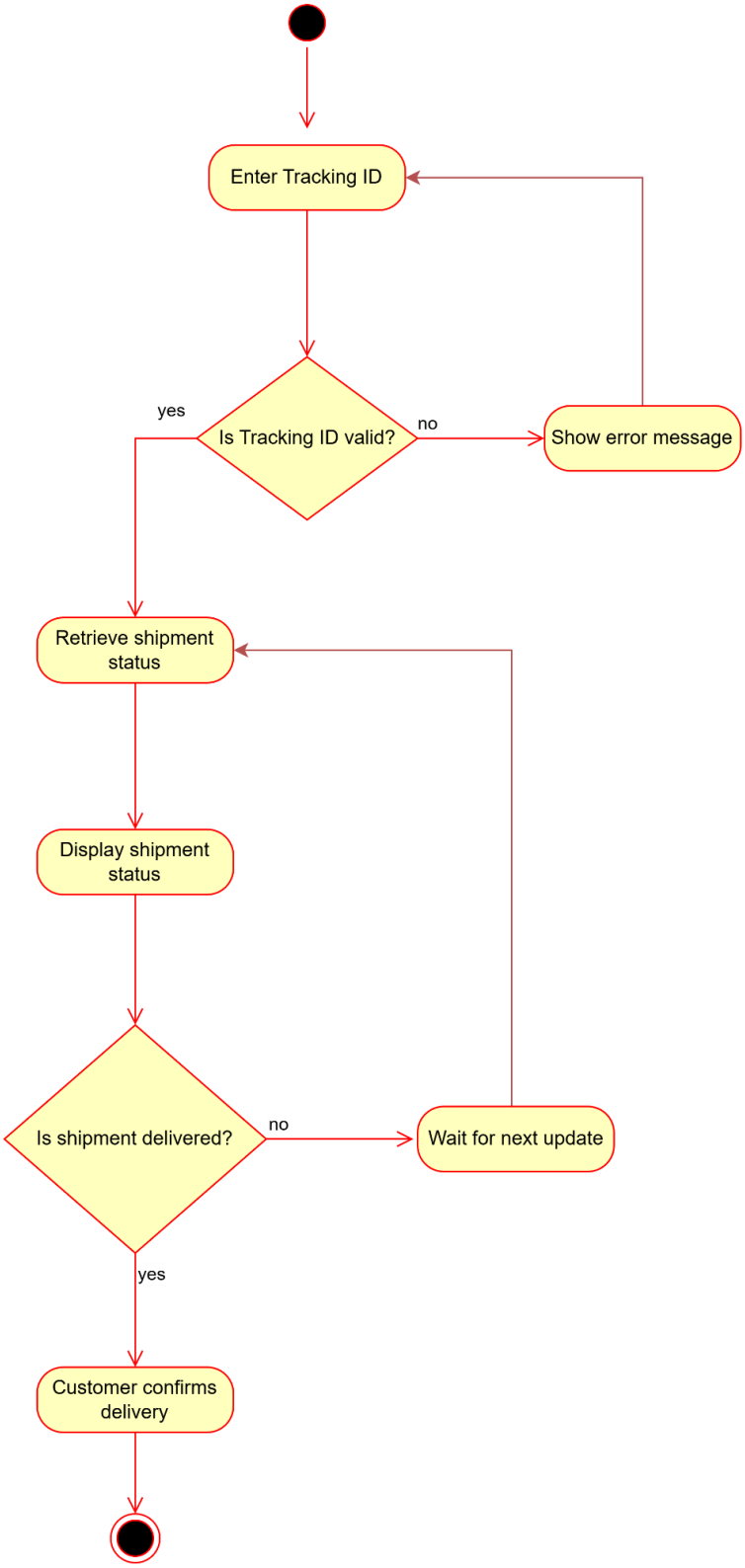
1. Order Management
2. Shipment Tracking
3. Delivery Execution
4. Exception & Return Handling
5. Shipment Tracking & Notification
6. Feedback & Complaint Management
7. Reporting & Monitoring
8. System Administration & Security

These diagrams help in understanding the control flow, validations, and exception scenarios involved in real-world delivery processes.

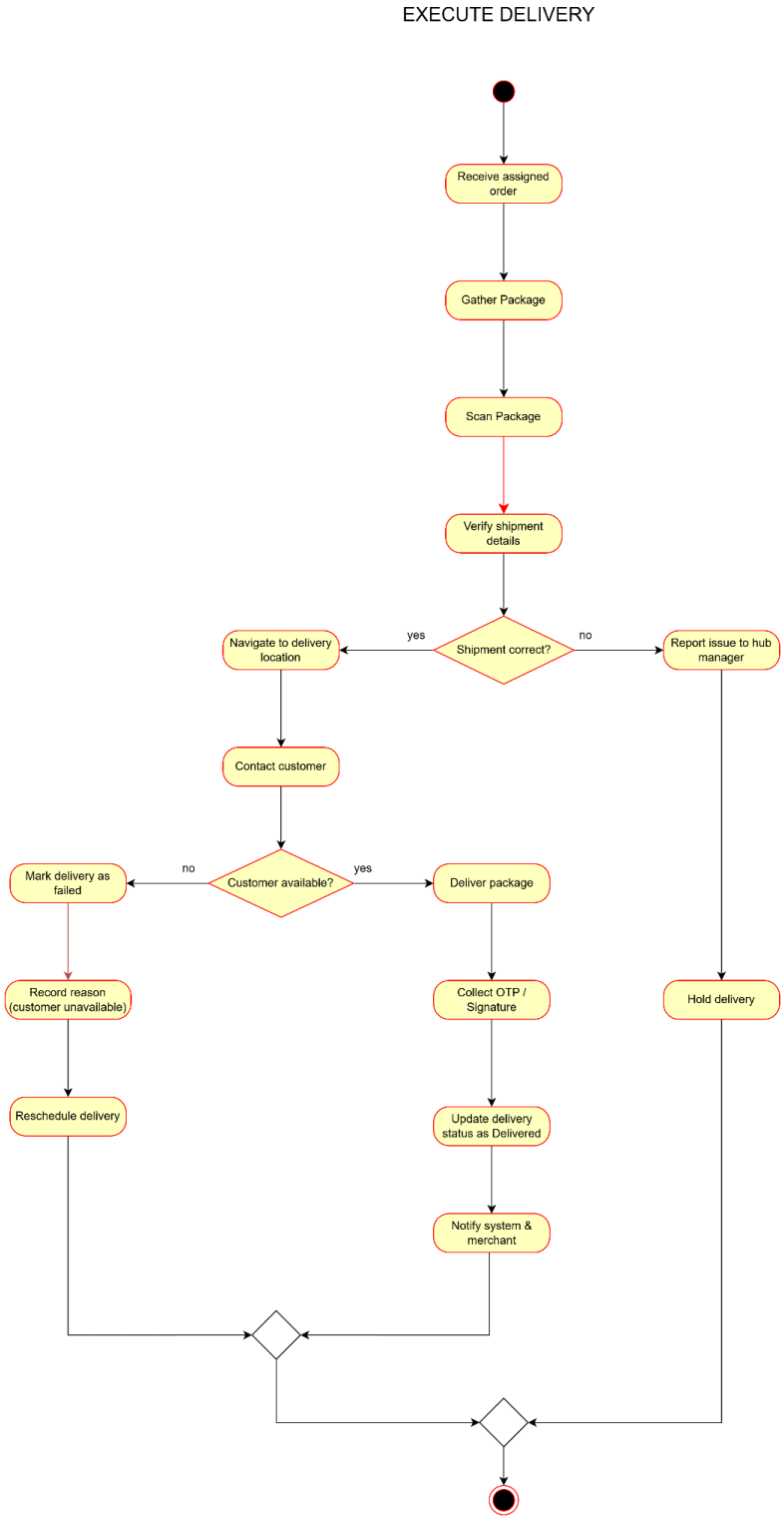
1. Order Management



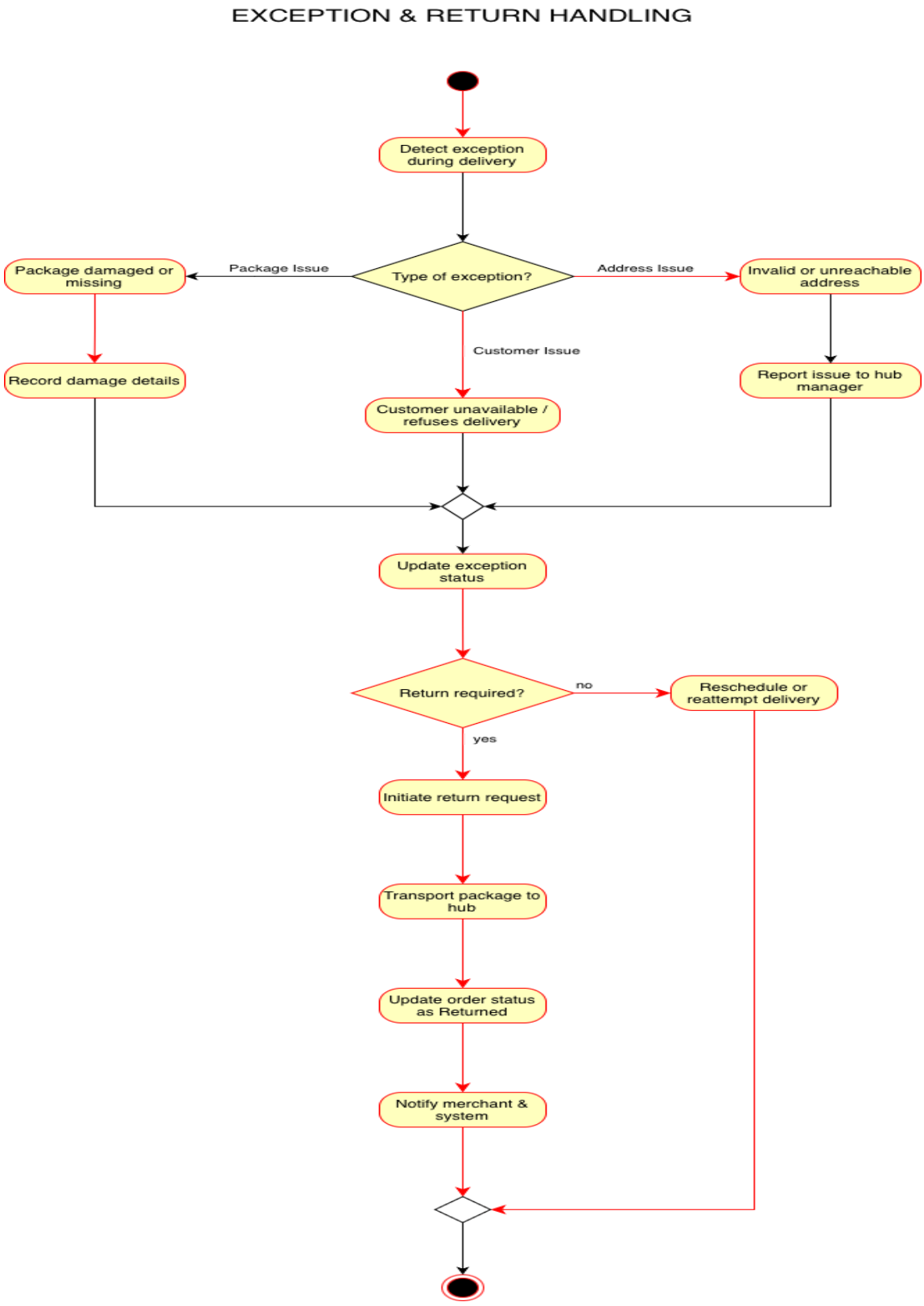
2. Shipment Tracking



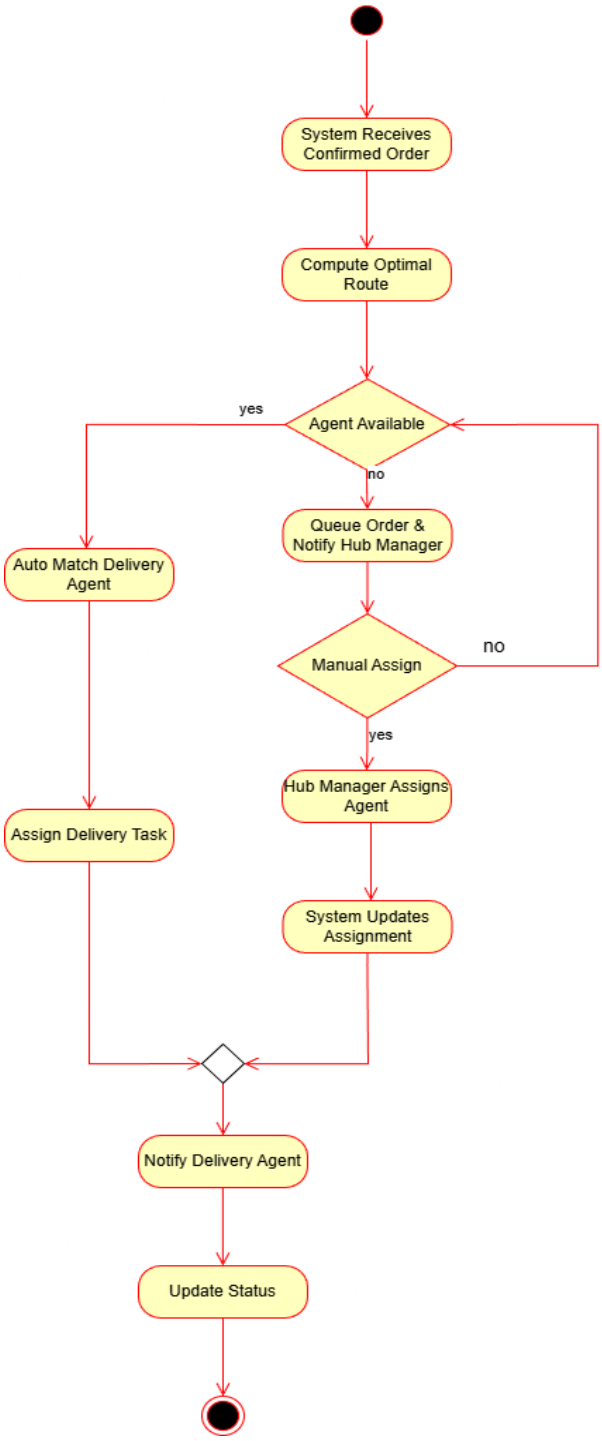
3. Execute Delivery



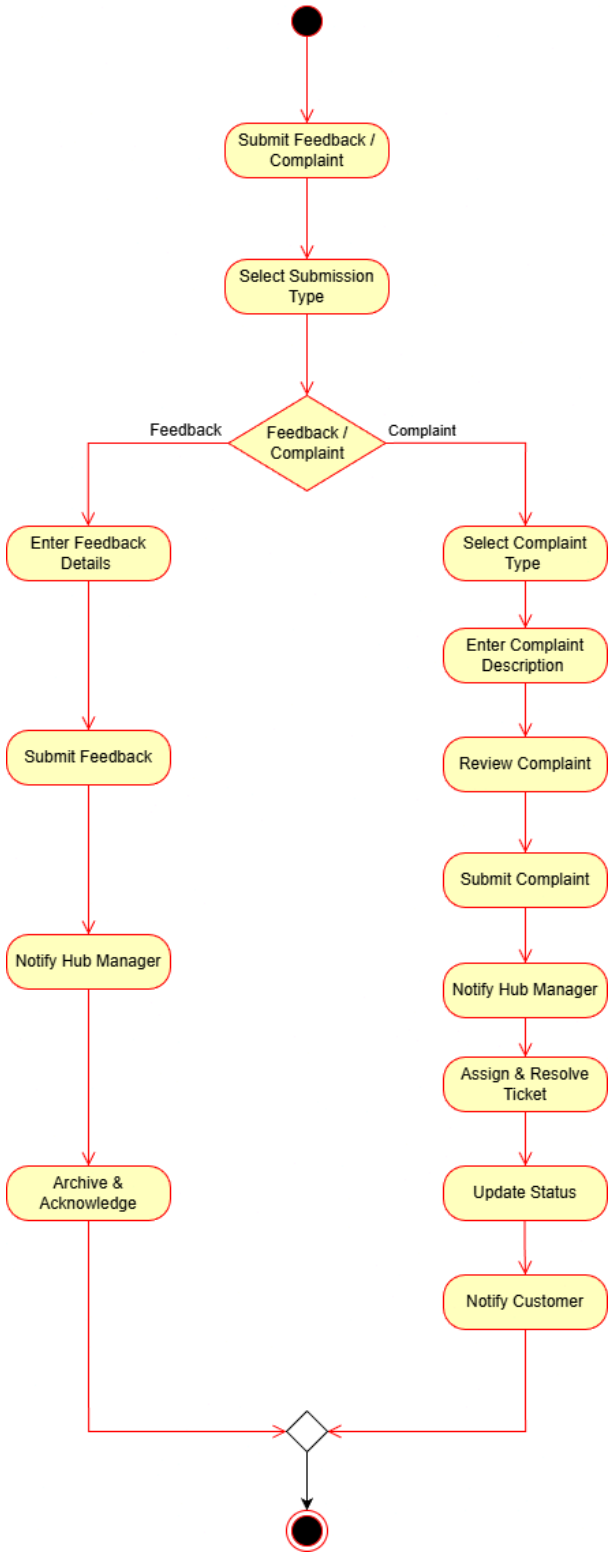
4. Exception & Return Handling



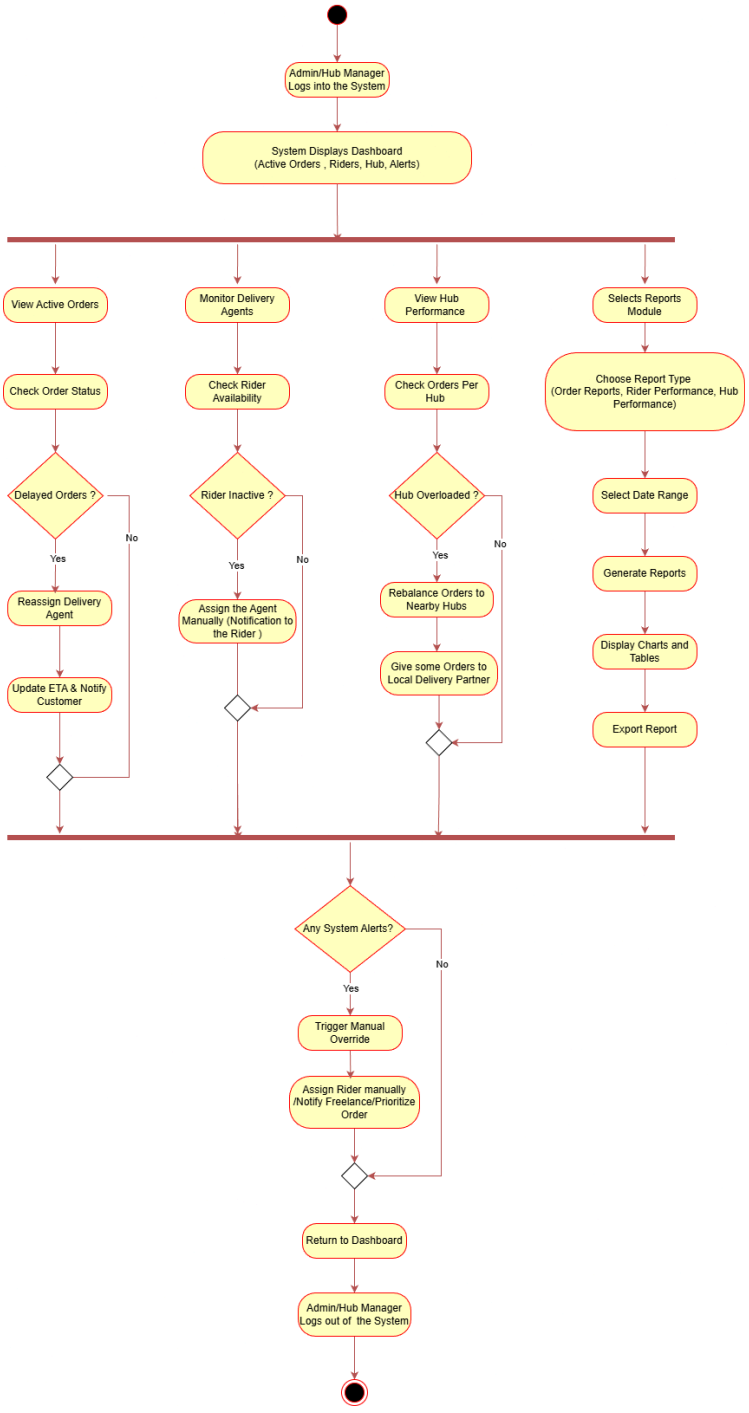
5. Dispatch & Assignment Management



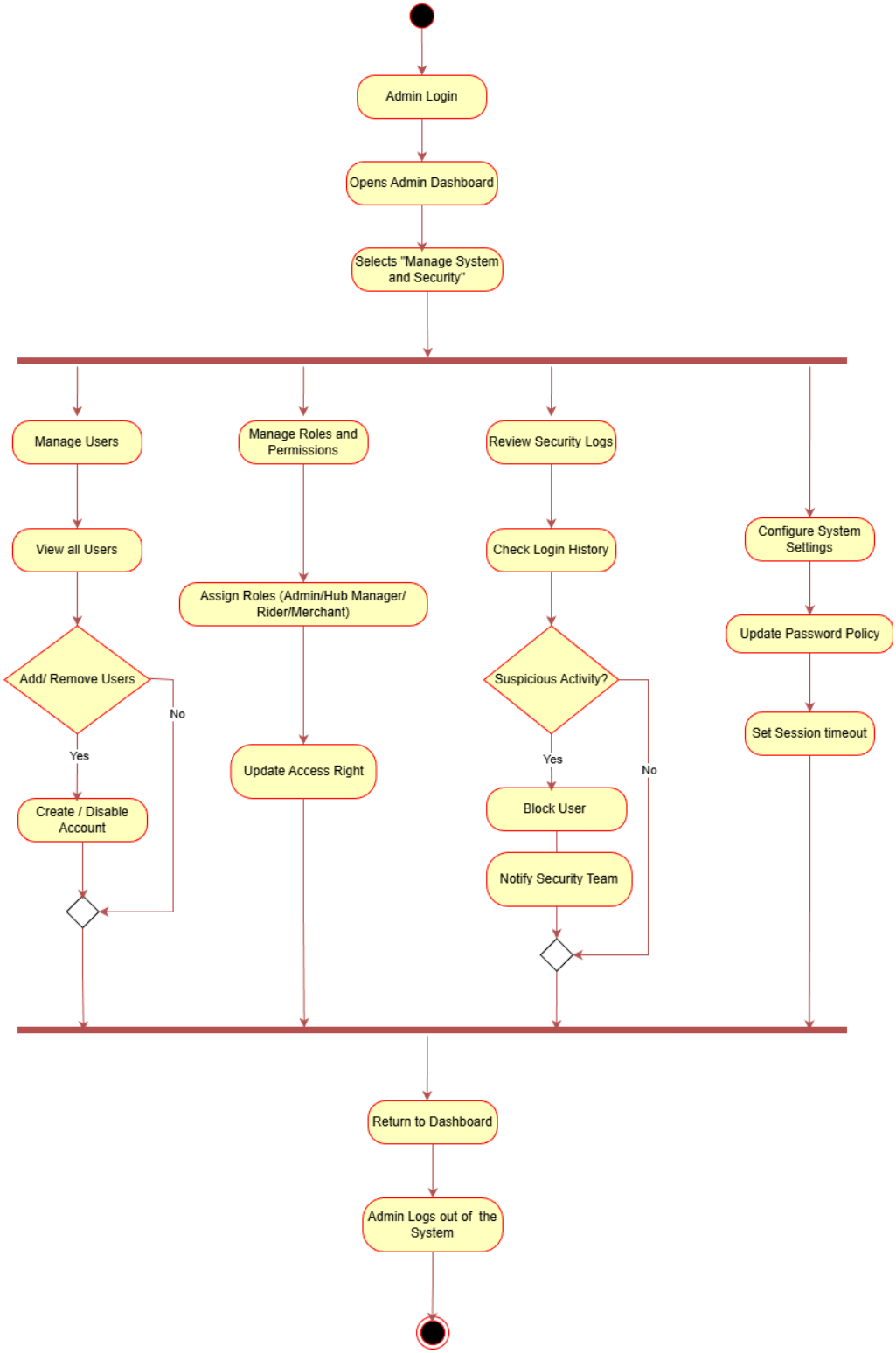
6. Feedback & Complaint Management



7. View Reports and Monitor Operations



8. System Administration & Security



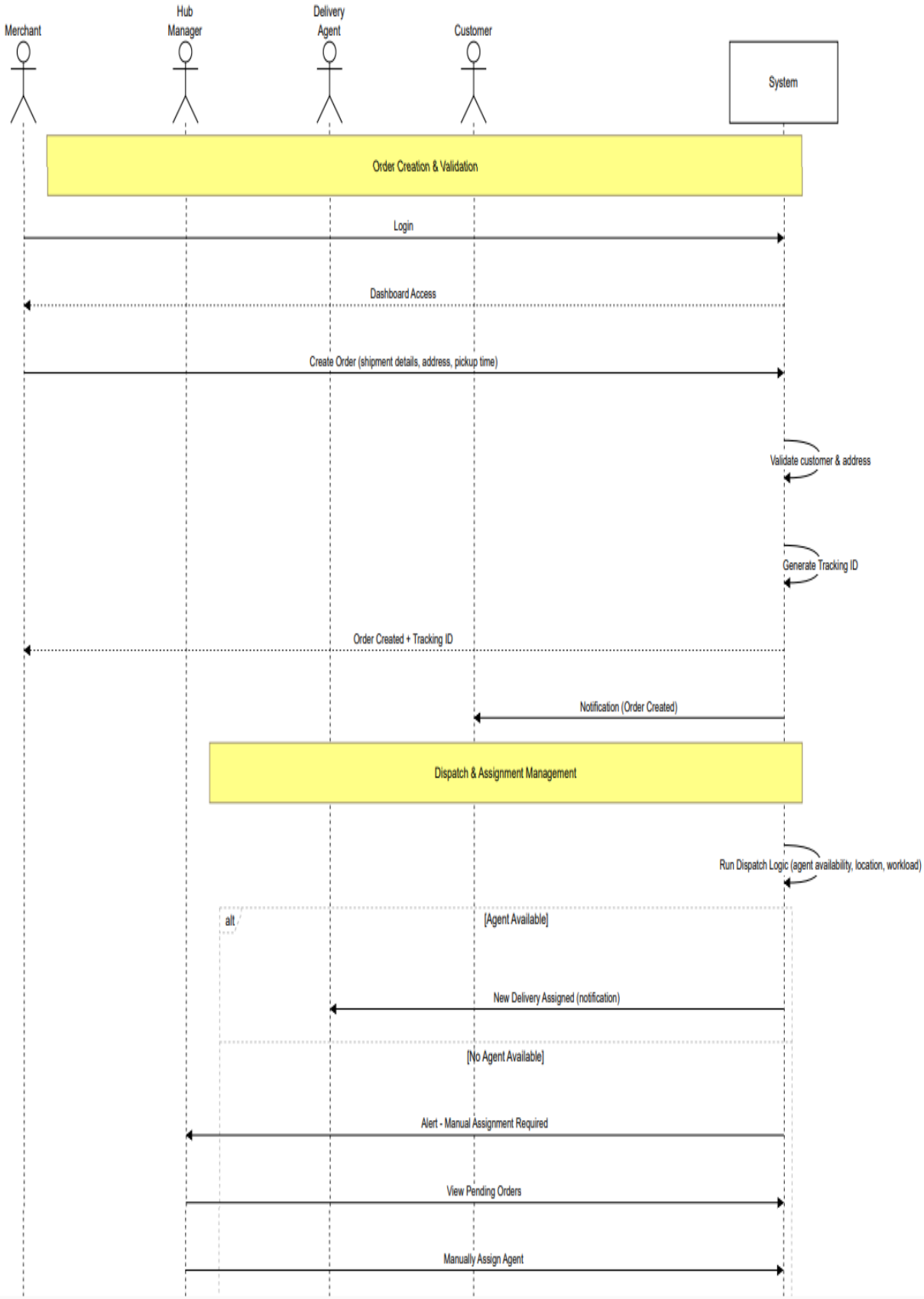
Sequence Diagram

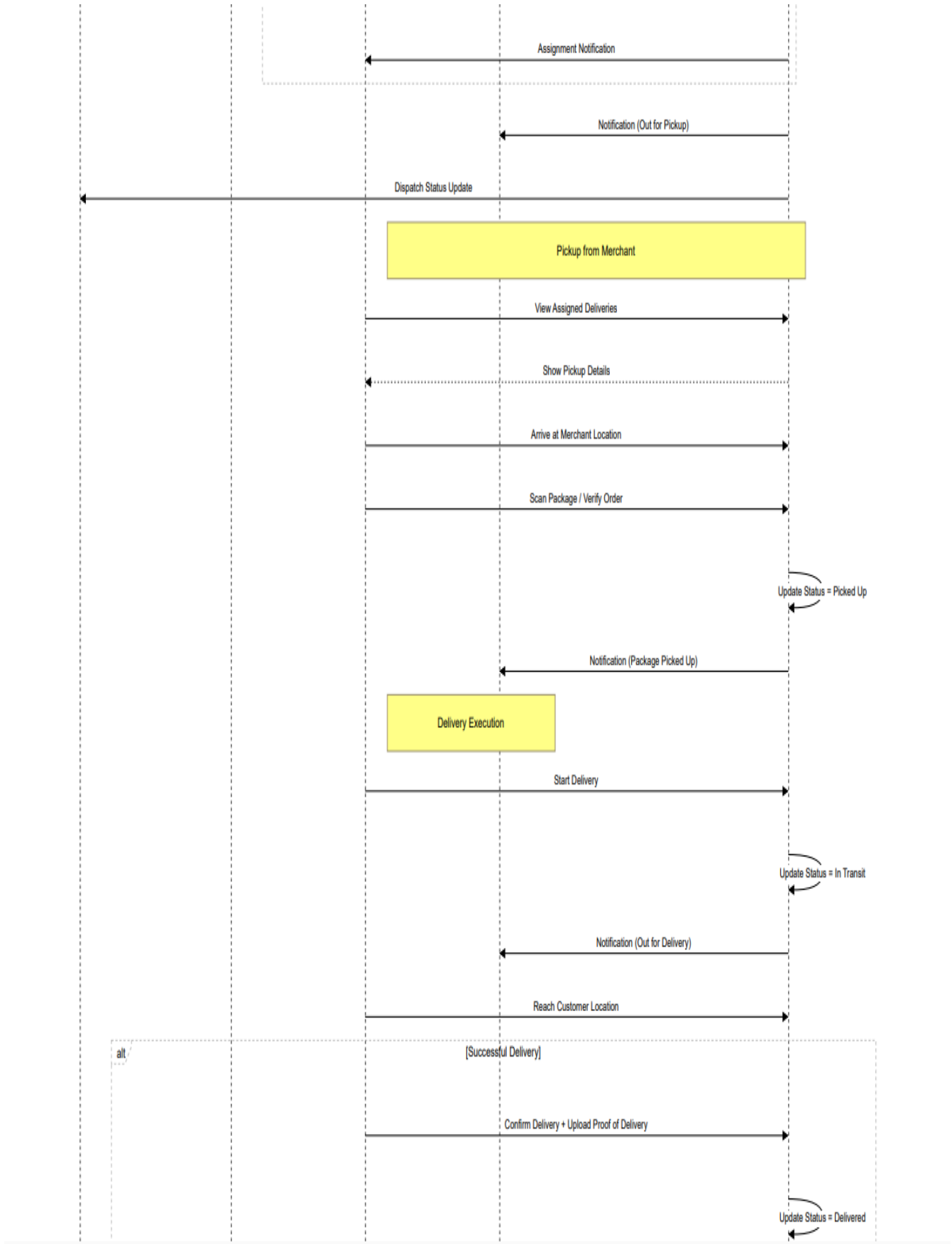
The Sequence Diagram represents the time-ordered interaction between actors and system components during the execution of key use cases.

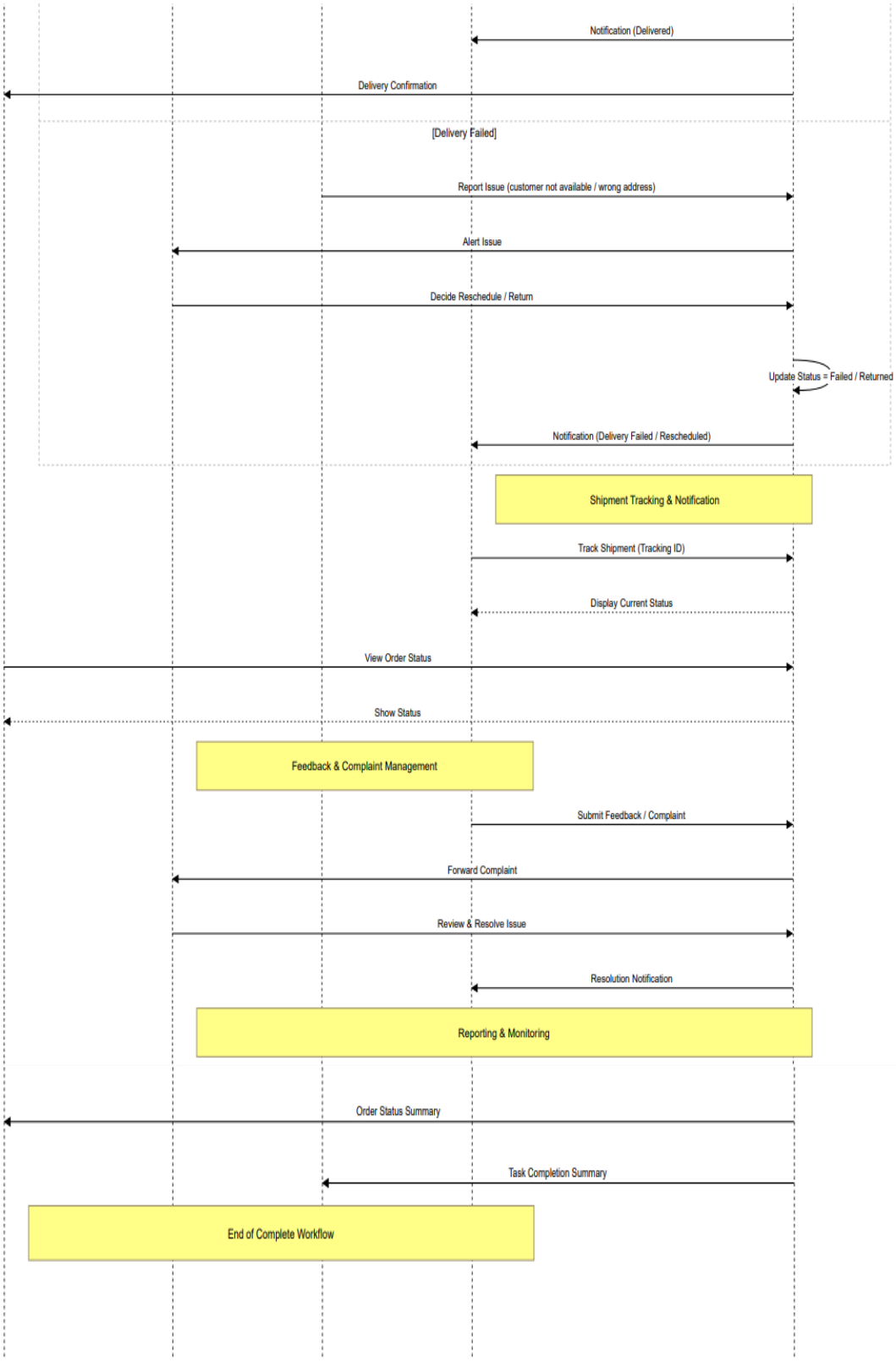
It demonstrates:

- How a merchant creates an order
- How the system assigns a delivery agent
- How the delivery agent updates delivery status
- How the customer tracks shipments and receives notifications
- How exceptions and feedback are handled

This diagram ensures that system interactions are logically ordered and aligned with the functional requirements.


Merchant
Hub Manager
Delivery Agent
Customer
System
Order Creation & Validation
Login
Dashboard Access
Create Order (shipment details, address, pickup time)
Validate customer & address
Generate Tracking ID
Order Created + Tracking ID
Notification (Order Created)
Dispatch & Assignment Management
Run Dispatch Logic (agent availability, location, workload)
alt
[Agent Available]
New Delivery Assigned (notification)
[No Agent Available]
Alert - Manual Assignment Required
View Pending Orders
Manually Assign Agent


Assignment Notification
Notification (Out for Pickup)
Dispatch Status Update
Pickup from Merchant
View Assigned Deliveries
Show Pickup Details
Arrive at Merchant Location
Scan Package / Verify Order
Update Status = Picked Up
Notification (Package Picked Up)
Delivery Execution
Start Delivery
Update Status = In Transit
Notification (Out for Delivery)
Reach Customer Location
alt
[Successful Delivery]
Confirm Delivery + Upload Proof of Delivery
Update Status = Delivered


Notification (Delivered)
Delivery Confirmation
[Delivery Failed]
Report Issue (customer not available / wrong address)
Alert Issue
Decide Reschedule / Return
Update Status = Failed / Returned
Notification (Delivery Failed / Rescheduled)
Shipment Tracking & Notification
Track Shipment (Tracking ID)
Display Current Status
View Order Status
Show Status
Feedback & Complaint Management
Submit Feedback / Complaint
Forward Complaint
Review & Resolve Issue
Resolution Notification
Reporting & Monitoring
Order Status Summary
Task Completion Summary
End of Complete Workflow